

Week-Wise Project Plan

Week	Tasks	Expected Outcome
Week 1	Study the problem domain: RAG, chunking, semantic search, multi-hop QA, hallucination issues. Read papers on RAG, DPR, RAPTOR, GraphRAG, semantic chunking, and HotpotQA.	Clear understanding of why fixed-size chunking and single-stage retrieval fail for long structured documents.
Week 2	Finalize project scope, objectives, research gaps, and system architecture. Prepare high-level architecture diagrams for CAAC and DR-RAG.	Finalized problem statement, objectives, architecture, and methodology skeleton.
Week 3	Set up development environment: Python, virtual environment, Git, Jupyter/VS Code, LangChain or LlamaIndex, embedding model, vector DB, evaluation libraries.	Working local development setup with basic document ingestion pipeline.
Week 4	Build baseline RAG pipeline using fixed-size chunking and single-stage retrieval. Use sample PDFs, academic papers, and legal text.	Baseline system ready for comparison against the proposed DR-RAG system.

Week 5	Implement document preprocessing: PDF/text extraction, cleaning, sentence splitting, metadata extraction, document type detection.	Clean structured input documents ready for adaptive chunking.
Week 6	Implement preliminary sentence-level embedding and semantic drift calculation between adjacent sentences.	First version of semantic velocity measurement.
Week 7	Design the SLM-guided boundary detection module. Create prompts/rules for academic papers, legal contracts, and general documents.	Initial Context-Augmented Agentic Chunking boundary detector.
Week 8	Implement adaptive chunk refinement: merge low-drift chunks, preserve moderate-drift boundaries, split high-drift sections.	Semantic chunks generated based on document meaning rather than fixed token size.
Week 9	Implement rolling-summary generation from previous chunks. Prepend summaries to current chunks before embedding.	Context-augmented chunks that preserve previous document context.
Week 10	Add token-budget scaling based on semantic velocity. Low-drift chunks get short summaries; high-drift chunks get longer summaries.	Final version of CAAC chunking pipeline.

Week 11	Store augmented chunks in vector database with metadata: document ID, section, chunk ID, velocity score, summary, raw chunk.	Searchable vector index with metadata filtering support.
Week 12	Implement query complexity classifier. Classify queries as simple/single-hop or complex/multi-hop using features like query length, conjunctions, semantic entropy, and model-based classification.	Query router that decides whether to use standard RAG or DR-RAG.
Week 13	Implement Stage 1 anchor retrieval using vector similarity search. Retrieve top-k chunks for every query.	Anchor retrieval module working for both simple and complex queries.
Week 14	Implement intermediate fact extraction from anchor chunks using SLM prompts or structured extraction. Extract entities, claims, relations, and key propositions.	Anchor chunks converted into useful intermediate facts.
Week 15	Implement mining query generation by combining original query with extracted anchor facts.	Expanded query generation module ready for second-stage retrieval.
Week 16	Implement Stage 2 evidence mining. Retrieve additional weakly linked supporting chunks and remove duplicates.	Full DR-RAG retrieval pipeline completed.

Week 17	Implement final answer generation using anchor + mined evidence. Add citations/source chunk references where possible.	End-to-end DR-RAG system producing grounded answers.
Week 18	Evaluation phase: compare baseline RAG vs CAAC + DR-RAG using Recall@K, MRR, Exact Match, latency, hallucination rate, and Context Retention Score.	Quantitative results showing improvement over baseline.
Week 19	Perform ablation study: remove rolling summaries, remove CAAC, remove query router, remove evidence mining. Analyze which component contributes most.	Ablation results and performance comparison tables.
Week 20	Final documentation, report writing, diagrams, results, conclusion, limitations, and future scope. Prepare presentation/demo.	Final major project report, presentation, and working prototype/demo.